

USB CLI Reference

MyWota (Pico water dispenser)

Volume-based water dispenser firmware on RP2040. Uses ESP8285 for WiFi (if enabled), MIFARE for prepaid balance in millilitres, and a local self-clean cycle. Adapter file:
application/Source/Adapters/USB_Command_Adapter.c

Source: sevantica_drivers/Source/USB/USB_Command_Handler.c (core) + project adapter
application/Source/Adapters/USB_Command_Adapter.c. Adapter commands are dispatched *before* core commands, so an adapter may shadow a core command of the same name.

help / ?

Origin: core

Show available top-level commands (adapter + core).

Command	Description	Example
help	List all registered commands with descriptions.	<code>help</code>
?	Alias of `help`.	<code>?</code>
clear	Clear the terminal screen and re-print the welcome banner.	<code>clear</code>

sys

Origin: core

System-level inspection and lifecycle control.

Command	Description	Example
sys status	Print device ID, site ID, RTC, MIFARE state, free heap, etc.	<code>sys status</code>
sys config	Dump the full live configuration (`Config_PrintToUSB`).	<code>sys config</code>
sys set <param> <value>	Set a single configuration parameter and persist to SD + flash.	<code>sys set mifare.card_init_default_balance 5000</code>
sys version	Show firmware version and build metadata.	<code>sys version</code>
sys reboot	Software reset via watchdog.	<code>sys reboot</code>
sys bootloader	Reboot into RP2040 BOOTSEL (USB mass-storage) mode for re-flashing.	<code>sys bootloader</code>
sys tasks	Print FreeRTOS task list with min-free stack (words).	<code>sys tasks</code>
sys wdt [save erase raw]	Watchdog log: dump latest entries; `save` flushes RAM->flash, `erase` wipes the WDT sector, `raw` dumps the raw flash slots.	<code>sys wdt</code> <code>sys wdt save</code> <code>sys wdt erase</code>

Command	Description	Example
sys security <on off>	Enable / disable the full MIFARE security stack (encryption, HMAC, replay, challenge-response, custom sector keys).	sys security on
sys factory-reset	Reload immutable defaults from the flash factory sector.	sys factory-reset
sys factory-save	Snapshot current source defaults into the flash factory sector.	sys factory-save
sys defaults-apply	Overwrite live (user) settings with the immutable source defaults.	sys defaults-apply

mod

Origin: core

Runtime control of system modules (start / stop / list).

Command	Description	Example
mod list	Print module table with current run state.	mod list
mod start <module>	Start a module. Known names: lcd, mifare, dispenser, buzzer, ioexp, rs485.	mod start dispenser
mod stop <module>	Stop a module by name (same names as `mod start`).	mod stop buzzer

card

Origin: core | Build flag: USE_DRIVERS_NFC

MIFARE card maintenance and recovery (gated by USE_DRIVERS_NFC).

Command	Description	Example
card format	Initialize / re-format the next presented MIFARE card with default data.	card format
card topup <amount>	Add tokens / volume units to the next presented card.	card topup 5000
card log <UID_HEX>	Read and print the on-card transaction log for the given UID.	card log 04A1B2C3D4
card recover <UID_HEX>	Restore card balance from the SD-side recovery log for that UID.	card recover 04A1B2C3D4
card decrypt	Decrypt and dump the next presented card (10 s wait) using current keys.	card decrypt

fs

Origin: core | Build flag: USE_DRIVERS_FATFS

SD card file system access (gated by USE_DRIVERS_FATFS).

Command	Description	Example
fs ls	List files in the SD root and cache them for `fs cat #N` access.	<code>fs ls</code>
fs cat <name #N>	Print contents of a file by name, or by index from the last `fs ls`.	<code>fs cat config.txt</code> <code>fs cat #2</code>
fs find <UID_HEX>	Search SD logs for entries belonging to that card UID.	<code>fs find 04A1B2C3D4</code>
fs format	Delete every file on the SD card (does NOT low-level format).	<code>fs format</code>
fs mount	Expose the SD card to the PC via USB Mass Storage; suspends application I/O.	<code>fs mount</code>
fs unmount	Detach USB MSC and resume application access to the SD card.	<code>fs unmount</code>

wifi (core / ESP8285)

Origin: core | Build flag: USE_DRIVERS_CONNECTIVITY

WiFi via the ESP8285 driver (gated by USE_DRIVERS_CONNECTIVITY).

Command	Description	Example
wifi scan	Trigger an asynchronous scan; results in 3-5 s.	<code>wifi scan</code>
wifi list	Print SSID / RSSI / security from the last scan.	<code>wifi list</code>
wifi status	Show driver state and IP address.	<code>wifi status</code>

rtc

Origin: core

Real-time clock inspection and manual set.

Command	Description	Example
rtc status	Print current time, sync flag, unix timestamp and compile-time fallback.	<code>rtc status</code>
rtc set <YYYY-MM-DD> <HH:MM:SS>	Set RTC date/time and persist to SD.	<code>rtc set 2026-05-04 14:30:00</code>

dispenser

Origin: adapter

Volume-based water dispenser control (MyWota adapter).

Command	Description	Example
dispenser start [L]	Start a manual dispense. Without arg -> unlimited; with arg -> N litres (min 1 L, max 1000 L).	<code>dispenser start</code> <code>dispenser start 5</code>

Command	Description	Example
dispenser stop	Stop an in-progress manual dispense.	<code>dispenser stop</code>
dispenser topup <L>	Top up the next presented card by N litres (min 1 L). Internally multiplied to mL.	<code>dispenser topup 10</code>

clean

Origin: adapter

Local self-clean cycle for the dispenser (MyWota adapter).

Command	Description	Example
clean start [vol_ml] [max_sec]	Start a self-clean cycle. Missing/zero args use the configured defaults.	<code>clean start</code> <code>clean start 2000 60</code>
clean stop	Abort a running self-clean cycle.	<code>clean stop</code>
clean status	Print whether a clean is active and the last-clean unix timestamp.	<code>clean status</code>

fault

Origin: adapter

Inspect and clear the dispenser fault state machine (MyWota adapter).

Command	Description	Example
fault status	Print current fault state, reason and incident counter (since boot).	<code>fault status</code>
fault clear	Clear a latched FAULT (operator action).	<code>fault clear</code>

Missing / Inconsistent Commands

Cross-checked against: BigYellow (Pico car-wash terminal), Central Control Hub (Pico 2 W master).

`filter` — missing in MyWota (Pico water dispenser)

Present in: Central Control Hub (Pico 2 W master).

`remote` — missing in MyWota (Pico water dispenser)

Present in: Central Control Hub (Pico 2 W master).

`rs485` — missing in MyWota (Pico water dispenser)

Present in: Central Control Hub (Pico 2 W master).

`sdinit` — missing in MyWota (Pico water dispenser)

Present in: Central Control Hub (Pico 2 W master).

`stats` — missing in MyWota (Pico water dispenser)

Present in: Central Control Hub (Pico 2 W master).

`washstart` — missing in MyWota (Pico water dispenser)

Present in: BigYellow (Pico car-wash terminal).

`washstop` — missing in MyWota (Pico water dispenser)

Present in: BigYellow (Pico car-wash terminal).

`webserver` — missing in MyWota (Pico water dispenser)

Present in: Central Control Hub (Pico 2 W master).

`stats` — missing

CCH exposes per-slave throughput buckets via `stats`. MyWota has no equivalent for its local dispenser (would need a Throughput_Stats adapter).

`filter` — missing

MyWota has its own filter but no CLI surface. Site-side filter accounting only exists on CCH. A local `filter status|reset|set` would parallel CCH.

`webserver` — missing

ESP8285 wifi has no HTTP server hook. If a config/diag UI is wanted, a wrapper around an ESP-side server would be required.

`wifi connect/disconnect/saved/forget/reinit` — missing

Core `wifi` exposes only `scan/list/status` for ESP8285. No way to join a network from the CLI without using `sys set` on raw config keys.